

Name: _____

ALGEBRAIC EXPRESSIONS, (EXPANDING SINGLE BRACKETS)

Date: _____

LO: I can expand single brackets by multiplying every term inside by the term outside.

What does 'expand' mean?

Expanding means removing the brackets by multiplying **every term** inside by the term outside.

Common error: only multiplying the first term. You must multiply BOTH terms inside the bracket.

Examples

●	$3(x + 2) = 3x + 6$	both terms
●	$-2(x - 5) = -2x + 10$	watch signs
●	$x(x + 3) = x^2 + 3x$	$x \times x = x^2$
●	$3(x + 2) = 3x + 2$	missed second term
●	$-2(x + 3) = -2x + 6$	should be -6

Key idea

Multiply the term outside by EVERY term inside. Watch the signs carefully when there's a negative outside.

$$3(x + 2) = 3x + 6$$

Multiply both terms

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - all positive

Expand $3(x + 4)$

$$3 \times x = 3x$$

$$3 \times 4 = 12$$

Combine: $3x + 12$

$$= 3x + 12$$

Multiply each term inside by 3.

Example 2 - with subtraction

Expand $5(x - 2)$

$$5 \times x = 5x$$

$$5 \times (-2) = -10$$

Combine: $5x - 10$

$$= 5x - 10$$

The minus sign belongs to the 2.

Example 3 - negative outside

Expand $-2(x + 3)$

$$-2 \times x = -2x$$

$$-2 \times 3 = -6$$

Combine: $-2x - 6$

$$= -2x - 6$$

Negative $\times$ positive = negative.

Example 4 - variable outside

Expand $x(x + 5)$

$$x \times x = x^2$$

$$x \times 5 = 5x$$

Combine: $x^2 + 5x$

$$= x^2 + 5x$$

x times x is x squared.

YOUR TURN – PRACTICE (deliberate practice)

1. Number outside, all positive

Multiply each term inside.

- a) $2(x + 5)$
- b) $4(x + 3)$
- c) $7(x + 2)$
- d) $5(x + 6)$

2. With subtraction inside

The minus sign stays with its term.

- a) $3(x - 4)$
- b) $6(x - 1)$
- c) $8(x - 5)$
- d) $9(x - 2)$

3. Negative outside

Watch signs carefully.

- a) $-2(x + 3)$
- b) $-5(x + 4)$
- c) $-3(x - 2)$
- d) $-4(x - 7)$

4. Variable outside

$x \times x = x^2$.

- a) $x(x + 4)$
- b) $x(x - 3)$
- c) $2x(x + 5)$
- d) $3x(x - 2)$

5. Mixed expand and simplify

Expand, then collect any like terms.

- a) $3(x + 2) + 4$
- b) $5(x - 1) + 2x$
- c) $2(x + 3) + 3(x - 1)$
- d) $x(x + 2) - x$
- e) $-2(x - 4) + 5$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $3(x + 2) = 3x + 2$ ☐

Correct answer: _____

Reason: _____

b) $5(x - 3) = 5x - 15$ ☐

Correct answer: _____

Reason: _____

c) $-4(x + 2) = -4x + 8$ ☐

Correct answer: _____

Reason: _____

d) $x(x + 3) = x^2 + 3x$ ☐

Correct answer: _____

Reason: _____

e) $-2(x - 5) = -2x + 10$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangle has length $(x + 4)$ cm and width 3 cm. Write an expanded expression for its perimeter and area.

Expression: _____

Simplified: _____

b) Expand and simplify $4(2x + 3) - 2(x - 5)$. Show your full working step by step.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

